INF221 - [Modern] Web Design & Development

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Learning Outcomes

By the end of this lecture, students will be able to:

- ❖ Analyze the structure of websites and web applications; Web Page Structure and Layers
  - ➢ **Structure**
  - ➢ **Presentation**
  - ➢ **Behavior**

Indicative Content

- ❖ Introduction
- ❖ The DOM
- ❖ Goals and Importance
- ❖ Standard Organisations

Introduction - Web standards

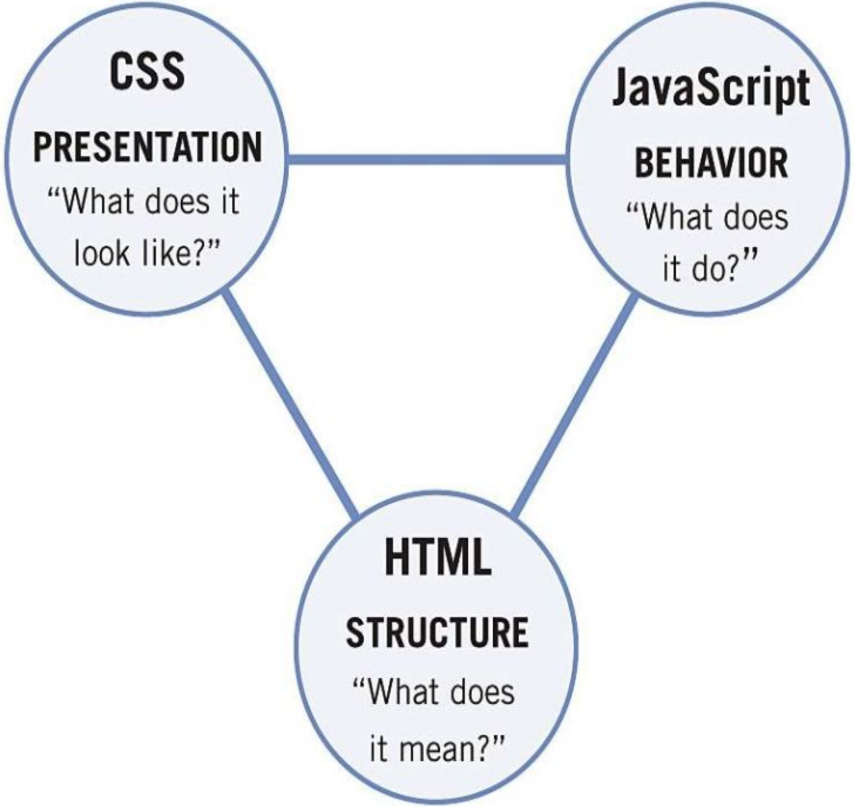
- ❖ Core **set of rules** for developing websites or web applications that may be accessed across different device configurations & environments
- ❖ Technologies used to create and interpret web-based content – established by standards bodies like the W3C (others???).
- ❖ They are designed (goals):
  - ➢ to future-proof documents published on the Web
  - ➢ to enable the accessibility of documents to as many people and devices as possible.
  - ➢ to enable developers or engineers design applications that are **functional**, **accessible** and **compatible** across various computer environments

Web standards…

- World Wide Web Consortium (W3C) is the main central body behind the setting of these standards
  - **Structural** – HTML, XHTML, XML
  - **Presentation** – CSS (Level 1, 2, 3 etc.), SVG (Scalable Vector Graphics)
  - **Behavior** – defined by the ECMAScript(ES), Describes a general purpose or standardized version of a programming language - JavaScript language
  - Document Object Model – DOM (Core)

The DOM - Core

The 3 technologies (HTML, CSS, JavaScript) **extend** or **build on top** of the CORE document object model.



Document Object Model (DOM)

- ❖ It is an application programming interface (API) for HTML & XML documents
  - ➢ Defines the logical structure of documents – specifies how they may be accessed and manipulated
  - ➢ The DOM is designed to be used with any web programming language – language independent
  - ➢ Abstract interfaces are specified for accessing and manipulating the internal representations of a document
- ❖ The main objective is to provide a standard programming interface
  - ➢ for usage in various environments & applications – W3C

Document Object Model (DOM)...

- Content - (X)HTML - Structure
  - Defines the content &/or structure of a document as specified by the DOM
  - Mandatory parts of the document – HTML. Semantic & Structured markup of a document
  - Each & every web browser validates this structure before it is rendered
  - Extensible HTML has its rules more stricter than HTML. It is a rewrite of HTML as an XML language
    - i.e. case sensitivity, closing of tags

Document Object Model (DOM)...

- ❖ Content - (X)HTML - Structure
  - ➢ XML
    - ■ “meta-language” – used to create & define other markup languages
    - ■ Examples GML, RSS
  - ➢ Since many of these markup languages were written in XML
    - ■ their contents can be shared across various applications that are using variants/implementations of XML

Web Standards - Importance

- Accessibility & Adaptation
- SEO – Search Engine Optimizations
- Backward compatibility
- Separation of concerns

Standard Bodies

- **W3C** - general web technology specifications body, created by TimBL (Tim Berners-Lee, 1990)
- **WHATWG** - maintains the living standards for the HTML
- **ECMA** - for ECMAScript (JavaScript standardisation)
- **Khronos** - publish technologies for 3D graphics, such as WebGL

References

❖ The web and web standards - Learn web development | MDN
❖ WhatWG
❖ ECMAScript® 2025 Language Specification